Chapter 25: INT 15H, BIOS extension functions

Use

Provides access to additional **Hunter 16**-specific functions via BIOS. The facilities are as follows:

60H Set clock speed
61H Get power level (APM)
62H Set low power threshold (APM)
63H Get/set idle timeout
64H Control Resume mode
65H Authorise charging (APM)
66H Set battery level (APM)
67H Control STOP mode

68H Power down
69H Control power down
6AH Control SLOW mode
6BH Get ROM BIOS version number
6CH Get serial number
6DH Get event details
6EH Enable/trap event
6FH Acknowledge event

70H Control sound
71H Power up keys
72H Redirect serial number
73H Get Oracle GT power level
74H Set backlight power up state
75H Set contrast power up state
76H Control power save
77H Reserved

78H Reserved
79H Redirect LPTI
7AH Invoke HOT key
7BH Control HOT key
7CH Control HOT key power option
7DH Override power warnings
7EH Control battery change facilities (APM)
7FH Set battery type (APM)

80H	Set battery capacity (APM)
81H	Control power output
82H	Sound output
83H	Control screen synchronisation
84H	Get/set diskette port
85H	Set/restore power menu (APM)
86H	Set attribute table
87H	Set INT 72H vector
88H	Get power up keys
89H	Get/set power mode
8AH	Control power input (SPM)
8BH	Set charger temperature override
8CH	Get/set power save entry flag
8DH	Get/set boost charge (SPM)

Note: functions marked (APM) only apply to Advanced Power Mode. Functions marked (SPM) only apply to Standard Power Mode.

Entry

AH selects the function required, while other registers are used for parameters.

Exit

Most of these functions return a status byte in AH, indicating success or failure of the operation:

AH	**Operation**
0	Successful
FF	Failed

Comments

INT 15 was used in the original IBM PC to implement cassette tape facilities (functions 0.3). In the IBM PC-AT and IBM PS/2 BIOS it is used to implement a variety of enhancements (functions 80..91H). The **Hunter 16** function numbers have been chosen so as not to clash with the published IBM enhancements.

Many of these functions are used to change the subsequent behaviour of **Hunter 16**; behaviour remains changed until the next cold start, which restores the default (normal) behaviour. (A cold start is produced by a power up in non-resume mode or after using emergency breakout or losing all power.)

Set clock speed (+)

Entry AH=60H
AL=mode

Exit Returns nothing

Description Use this function to alter execution speed, when lower power operation is required and execution speed is not critical.

AL	Speed
0	Normal (8 MHz)
1	Slow mode 1 (4 MHz)
2	Slow mode 2 (2 MHz)

The default, restored by a cold start, is 8 MHz clock speed.

Get power level (APM) (+)

Entry AH=61H

Exit AH=level

Description In Advanced Power Mode, use this function to read the power remaining, as a percentage of total battery power (ie 0..64H). (In Standard Power Mode, the battery level is not monitored and this function will always return 7FH.)

Set low power threshold (APM) (+)

Entry AH=62H
AL=level
BX=interval (1..0258H)

Exit AH=status

Description In Advanced Power Mode, **Hunter 16** will normally display the battery low power warnings every 5 seconds if the battery level falls to 5%, and turn off automatically after 10 warnings.

Use this function to change the power level at which warnings start, and the interval between warnings:

AL	Level
0..12H	5% to 95%

BX	Interval
1..258H	1 to 600 seconds

In Standard Power Mode, this function is ignored; low power warnings are issued when the battery voltage falls below a fixed threshold.

Get/set idle timeout (+)

Entry

AH=63H
AL=0 (get) or 1 (set)
BX=time (0..0E10H) if set

Exit

AH=status
BX=time (0.0E10H) if get

Description

Hunter 16 will normally turn off automatically if left idle (no keyboard input or communications activity) for 5 minutes.

Use this function to get or set the idle timeout:

BX	Timeout
0	Do not timeout
1..0E10H	1 to 3600 seconds

Control Resume mode (+)

Entry

AH=64H
AL=control

Exit

AH=status

Description

Hunter 16 will normally continue executing the current application when turned off and then turned on again, as if nothing had happened. This is 'resume mode'.

If resume mode is disabled, when **Hunter 16** is turned on it will display the 'Welcome' screen and boot MS-DOS. This is a cold start and will re-enable resume mode automatically (and set all other settings to their defaults).

Use this function to enable or disable resume mode:

AL	Effect
0	Disable resume mode
1	Enable resume mode

The default, restored by a cold start, is resume mode.

Authorise charging (APM) (+)

Entry

AH=65H
AL=charging level

BH=02H	(to
BL=03H	minimise
CH=04H	risk
CL=05H	of
DH=06H	inadvertent
DL=07H	operation)

Exit

AH=status

Description

Hunter 16 can be fitted with Alkaline batteries or NiCads. NiCads can be recharged, while Alkaline batteries must not.

In Advanced Power Mode **Hunter 16** does not permit recharging by default; when you connect the AC adaptor, it is simply used to power **Hunter 16** in place of any batteries. (In Standard Power Mode charging is enabled by default, but you can use function 8AH to disable it.)

Use this function to allow recharging by connecting the AC adaptor, IF YOU ARE SURE THAT RECHARGEABLE BATTERIES ARE FITTED. The charging level is indicated by AL;

the contents of the other registers must be as specified, to prevent inadvertent use of this function.

AL	Effect
1	Charging not allowed.
2	Charging allowed until power down (for any reason).
3	Charging allowed until batteries changed (or fail). Ask user if charging allowed when next turned on.
4	Charging allowed indefinitely. USE WITH CAUTION, AS USERS MAY FIT NON-RECHARGEABLE BATTERIES!

Changing the charging authorisation does enable charging directly; you must enable it by using function 7FH (or the user can use the HOT key power option). Changing the authorisation to 1 will however immediately disable charging.

Once charging is enabled, when the charger is connected, charging will adjust the power level automatically.

If the user changes batteries, their power level must be specified by the user (using the HOT key) or by function 66H.

Note that if all power is removed for long enough to drain the internal backup battery, the charging level will always be reset to 1 (no charging).

The user can also change the charging authorisation (and other battery characteristics) by using the HOT key power option, but you can prevent this by using function 7BH or 7CH. The user is also prompted to do this after changing batteries, but you can prevent this by using function 7EH.

Set battery level (APM) (+)

Entry

AH=66H
AL=level (0..64H)

Exit

AH=status

Description

In Advanced Power Mode (APM) **Hunter 16** monitors battery discharging (and charging, for NiCads) continuously, so that it can maintain an accurate measurement of the current battery level. (In Standard Power Mode, the battery level is not monitored.)

It also detects when the battery cap has been removed and will normally prompt the user to set the battery characteristics (including the level) when next turned on. This allows the user to specify the new batteries' characteristics when they have been changed. (Although you can suppress this by using function 7EH.)

You can also use this function to set the battery level directly, if you know that the batteries have been changed or have some other reason to believe that the current level is incorrect:

AL	Effect
0..64H	Level (0..100%)

Control STOP mode (+)

Entry

AH=67H
AL=mode

Exit

AH=status

Description

Hunter 16 normally enters STOP mode when wailing for keyboard input, lo conserve power while allowing communications to proceed normally.

Use this function to enable or disable STOP mode. With STOP mode disabled, HALT mode will be used instead. This conserves even more power,

but does not allow communications:

AL	Effect
0	Use STOP mode
1	Use HALT mode

The default, restored on a cold start, is STOP mode.

Power down (+)

Entry

AH=68H

Exit

Returns nothing

Description

If power down is allowed (see function 69H), this turns **Hunter 16** off. When Hunter 16 powers up, it will either boot MS-DOS (if not in resume mode), or return to the application (if in resume mode).

If power down is not allowed, it will return immediately.

For details of resume mode, see function 64H.

Control power down (+)

Entry

AH=69H
AL=mode

Exit

AH=status

Description

This function can be used to prevent power down due to pressing the [PWR] key, the power down function (68H) or expiry of the idle timeout (see function 63H).

It cannot prevent power down due to power fail or low power.

AL	Effect
0	Inhibit power down
1	Allow power down

Control SLOW mode (+)

Entry

AH=6AH
AL=mode

Exit

AH=status

Description

Use this function to enable or disable selection of SLOW mode:

AL	Effect
0	Inhibit SLOW mode
1	Allow SLOW mode

This would typically be used to stop **Hunter 16** using SLOW mode for critical periods when you require full speed processing.

The default, restored on a cold start, is to allow SLOW mode.

Get ROM BIOS version number (+)

Entry

AH=6BH

Exit

BH=version number (ASCII)
BL=release number (ASCII)
CH=minor release number (ASCII)
CL=minor release number (ASCII)

Description

Use this function to read the ROM BIOS version and release number.

Get serial number (+)

Entry

AH=6CH

Exit

BL=top 2 digits (BCD)
CX=least significant 4 digits (BCD)

Description

By default, each Hunter 16 has the same 6-digit serial number, but serial numbers can be stored

in RAM (see function 72H). These can then be used to identify the unit in which the program is running (eg for copy-protection purposes).

Use this function to read the serial number, which is stored as ASCII digits.

Get event details (+)

Entry

AH=6DH

Exit

AH=0 (success)
BX=event flags, bits 0..15
CX-event flags, bits 16.31

Description

The events mechanism provides an easy way to detect important events such as the RTC alarm going off or special keys being pressed.

Use this function to read the events status. Any enabled events which have occurred but not yet been acknowledged (see function 6FH) will be indicated by setting the appropriate bits in BX and CX (ie event 0 is bit 0, event 1 is bit 1, etc):

Event	Description
0	[PWR] pressed
1	Error on COM1
2	Error on COM2
3	Data received on COM1
4	Data received on COM2
5	RTC alarm
6	RI alarm

(Events are enabled by using function 6EH.)
For more details of events, see Chapter 41.

Enable/trap event (+)

Entry

AH=6EH
AL=software interrupt no.
BH=event number
BL=enable mask

Exit

AH=0 (success)

Description

The events mechanism provides an easy way to detect important events such as the RTC alarm going off or special keys being pressed.

Use this function to enable or disable an event, or to trap the event,

BH defines the event number to control.

BL specifies whether to enable or disable that event:

BL	Effect
0	Disable (ignore) event
1	Enable event

Enabling an event replaces its normal effect. For example, if event 0 is enabled, pressing the [Ctrl][C] key will not insert the corresponding key token in the keyboard buffer, so the BIOS will ignore the key entirely. Instead, the corresponding event flag will be set or the trap for that event will be executed.

Disabling an event restores its normal effect. For example, pressing [Ctrl][C] will insert the corresponding key token in the keyboard buffer, and will no longer set that event flag or execute the trap defined for that event.

AL specifies whether to trap the event:

AL	Effect
0	Do not trap
1..FFH	Trap as software interrupt 1..255

If you trap an enabled event, the corresponding software interrupt will be called when the event occurs. You must define the interrupt service routine (by using the MS-DOS Set Vector function) before trapping the event.

The interrupt service routine MUST NOT execute any MS-DOS or BIOS interrupts and must be short. Typically, it will set a flag within the application which the application polls at regular intervals.

By default, all events are disabled and no events are trapped. The defaults are restored on a cold start.

For more details of events, see Chapter 41.

Acknowledge event (+)

Entry

AH=6FH
AL=event number

Exit

AH=0 (success)

Description

The events mechanism provides an easy way to detect important events such as the RTC alarm going off or special keys being pressed.

Use this function to acknowledge an event when you are about to service it. This resets the corresponding bit in the event status bytes, so you can detect the next occurrence of this event.

For more details of events, see Chapter 41.

Control sound (+)

Entry

AH=70H
AL=0 (disable) or non-0 (enable)

Exit

Registers unchanged

Description

Use this function to control ALL sound output, whether instigated by an application or BIOS.

Power up keys (+)

Entry

AH=71H
BL=column (see below)

BH=row (see below)
AL=column switch (0=disable, 1=enable)
CL=row switch (0=disable, 1=enable)

Exit

AH=status

Description

Use this function to define one or more extra power up keys, by specifying their row and column numbers.

You can define as many keys as you wish. For example, by repeated calls, you could define ALL keys as power up keys, so any key depression would turn **Hunter 16** on.

The key row and column numbers are summarised in the following table.

Row	Col 0	Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7	Col 8	Col 9	Col 10
7	Esc	1	Q	Tab	Caps Lock	\	-	-	left Shift	-	-
6	2	W	A	S	Z	-	Alt	-	-	-	-
5	4	3	E	R	D	F	X	C	space	-	-
4	0	-	'	7	↵	4	.	1	0	-	-
3	+	←	8	9	5	6	2	3	DEL	-	-
2	9	O	P	;	L	,	-	right Shift	-	-	-
1	8	7	U	I	J	K	M	N	/	-	-
0	6	5	T	Y	G	H	B	V	#	-	-
	PWR	-	-	-	-	-	-	-	-	Ctrl	♠

(Keys shown like 1 are in the numeric keypad.)

Note that these keys will only power **Hunter 16**; they will not power down.

The default, restored by a cold start, is for only the [PWR] key to be a power up key.

Redirect serial number (+)

Entry

AH=72H
AL=0 (reset) or non-0 (set)
DS=segment address of serial number
BX=segment offset of serial number

Exit

AH=0 (success)

Description

By default, each **Hunter 16** has the same 6-digit serial number., but serial numbers can be stored in the RAM. These can then be used to identify the unit which a program is running on (eg for copy-protection purposes).

Serial numbers are read by using function 6CH.

Use this function to change the serial number read routine so that it returns the serial number stored in the RAM, or reset it to read the standard serial number.

The serial number should be stored as consecutive bytes, with each byte specifying a decimal digit in ASCII (ie in the range 30H (0) to 39H (9).

The default, restored by a cold start, is to use the Husky serial number (normally identical).

Get Oracle GT power level (+)

Entry

AH=73H
AL=drive (0 is A:, 1 is B:)

Exit

If AH=0 then AL=level:
25 decimal = 25%
50 decimal = 50%
75 decimal = 75%
100 decimal = 100%
If AH=FFH then the **Oracle GT** is not connected

Description Use this function to read the power remaining in the **Oracle GT** batteries, as a percentage of their total power.

Set backlight power up state (+)

Entry
AH=74H
AL=mode (1=off, 2=same level, 3=level as BL)
BL=level (0..7FH)

Exit AH=status

Description Use this function to specify how the backlight level should be set when **Hunter 16** is next turned on.

The default, restored by a cold start, is 2 (same level), but as the default level is 0 the backlight will be turned off.

Set contrast power up state (+)

Entry
AH=75H
AL=mode (1=off, 2=same level, 3=level as BL)
BL=level (0..7FH)

Exit AH=status

Description Use this function to specify the state how the LCD contrast should be set when **Hunter 16** is next turned on.

The default level, restored by a cold start, is 0.

Control power save (+)

Entry
AH=76H
BX=control

Exit Registers unchanged

Description **Hunter 16** enters power save modes under a variety of conditions. Use this function to control those conditions. BX defines the conditions. If bit

0 is 0, **Hunter 16** will not enter power save; if bit 0 is 1, power save is enabled but individual causes can be inhibited by setting bits 1 to 3.

Bit	
D0=0	Hunter 16 will not enter power save
D0=1	power save is enabled and following conditions apply:

Bit	**Inhibits power save:**
D1=1	If waiting for data from COM1
D2=1	If waiting for data from COM2
D3=1	If waiting for input from barcode wand
D4=1	Inhibit power up on timer tick

Hunter 16 will normally leave power save mode briefly about 18 times a second (each system clock "tick'), to increment the system clock counter. Set bit 4 to stop this and reduce power requirements still further, but note that this will make the system clock run slow.

By default, all power save causes are enabled. This default is restored by a cold start.

Redirect LPT1 (+)

Entry
AH=79H
AL=port

Exit
Returns nothing

Description
Redirects subsequent LPT1 output to COM1 if AL=0, or COM2 if AL=1.

Invoke HOT key (+)

Entry
AH=7AH

Exit
Returns nothing

Description

Invokes the HOT key function just as if the user had pressed the HOT key ([PAW][H]). Access is controlled by functions 7BH (inhibit all access) and 7CH (inhibit access to power functions), exactly as if the user pressed the HOT key.

Control HOT key (+)

Entry

AH=7BH
AL=control (0=prevent, non 0-allow)

Exit

Returns nothing

Description

The HOT key allows users to examine and change various aspects of **Hunter 16's** set-up. Use this function to prevent access to these facilities.

By default, access is enabled. This default is restored by a cold start.

If you only want to prevent access to the power control facilities (set power level, low power threshold, battery type, capacity and charging authorisation), use function 7CH.

Use of this function does not stop selection of the power option screen after a battery change

Control HOT key power option (+)

Entry

AH=7CH
AL=control (0=prevent, non 0=allow)

Exit

Returns nothing

Description

The HOT key power option allows users to switch between Standard Power Mode (SPM) and Advanced Power Mode (APM), and to set the power level, low power threshold, battery type, capacity and charging authorisation, in APM. Use this function to prevent access to these facilities while allowing normal use of the other HOT key menu options.

By default, access is enabled. This default is restored by a cold start.

If you want to prevent access to all the HOT key facilities, use function 7BH.

Override low power turn off (+)

Entry

AH=7DH
AL=control
(0=turn off after 10 warnings
Non-0=never turn off)

Exit

Returns nothing

Description

Hunter 16 will normally turn off after issuing 10 consecutive low power warnings. Use this function to disable this automatic power down, while still displaying the warnings. This allows prolonged used in emergency situations.

By default, power down is enabled. This default is restored by a cold start.

In Advanced Power Mode, you can alter the low power threshold and interval between warnings by using function 62H; a user can also alter the threshold by using the HOT key. In Standard Power Mode, low power warnings are issued if the battery voltage falls below a fixed threshold.

Control battery change facilities (APM) (+)

Entry

AH=7EH
AL=control 0 (do not prompt) or non 0 (prompt)

Exit

Returns nothing

Description

In Advanced Power Mode, if the user removes and replaces the battery cap, when **Hunter 16** turns on again, it will normally prompt for information about the batteries (by displaying the HOT key power option screen see Part 2). This allows the user to easily change the battery level,

capacity and charging authorisation after changing a battery. (In Standard Power Mode, this mechanism does not operate.)

Use this function to disable this mechanism, so that **Hunter 16** will simply power up normally. The user can still access these same facilities via the HOT key unless this has been disabled by using function 7BH or 7CH.

An application can also control the power settings directly, by using functions 65H (charging authorisation), 66H (level), 7FH (enable charging) and 80H (capacity). By default, **Hunter 16** will prompt for the information. The default is restored by a cold start.

Set battery type (APM) (+)

Entry

AH=7FH
AL=battery type
(0=non-rechargeable
non 0=rechargeable)

Exit

Returns nothing

Description

In Advanced Power Mode, use this function to enable or disable recharging. You can only enable charging if the charger authorisation level is higher than 1 (see function 65H). (In Standard Power Mode, charging is controlled by function 8AH.)

The user can also enable or disable recharging by using the HOT key power option (but see function 7CH) and after changing batteries (but see function 7EH).

The default battery type selected when you first switch to Advanced Power Mode is non-rechargeable. On a cold start, Standard Power Mode is selected.

Set battery capacity (APM) (+)

Entry

AH=80H
AL=0 (get) or non 0 (set)
BL=capacity (0..28H) if set

Exit

AL=capacity (0..28H) if get

Description

In Advanced Power Mode, use this function to get or set the battery capacity directly (ignored in Standard Power Mode):

BL	Capacity
0	500
1	550
2	600
...	
28H	2500

The maximum battery capacity is 2500mAh. The default capacity is 500mAh, which is restored by a cold start.

Control power output (+)

Entry

AH=81H
AL=0 (off) or non 0 (on)

Exit

Returns nothing

Description

Use this function to turn on or off the +5V low power output, provided on pin 10 of COM1 (see Appendix E for further details).
The power output will turn off when **Hunter 16** is turned off, but turn on again when **Hunter 16** is turned on.

The default state is off. This default is restored by a cold start.

Sound output (+)

Entry

AH=82H
BX=pitch
DX=length

Exit

Returns nothing

Description

Use this function to sound a tone of a specified pitch (BX) and duration (DX). These parameters are summarised in the following table.

Sound interrupt pitch parameters

BX (dec)	note	frequency (dec)	BX (dec)	note	frequency (dec)
4803	C	138.810	1430	A#	466.162
4668	C#	142.827	1350	B	493.880
4540	D	146.830	1274	C	523.250
4286	D#	155.560	1203	C#	554.365
4045	E	164.810	1135	D	587.330
3818	F	174.610	1071	D#	622.257
3604	F#	184.996	1011	E	659.260
3401	G	196.000	954	F	698.460
3210	G#	207.654	901	F#	739.990
3030	A	220.000	850	G	783.990
2860	A#	233.081	803	G#	830.609
2700	B	246.940	758	A	880.000
2548	MID C	261.630	715	A#	932.329
2405	C#	277.183	675	B	987.770
2270	D	293.660	637	C	1046.500
2143	D#	311.127	601	C#	1108.749
2022	E	329.630	568	D	1174.700
1909	F	349.230	536	D#	1244.523
1802	F#	369.998	506	E	1318.500
1701	G	392.000	477	F	1396.900
1605	G#	415.307	451	F#	1479.503
1515	A	440.000	425	G	1568.000

The length parameter, DX, can be calculated from the following formula:

$$DX = \frac{duration * 666670}{frequency}$$

For example, to sound middle C for 0.5 seconds, set DX to 1274 decimal and BX to 2548 decimal.

Control screen synchronisation (+)

Entry

AH=83H
AL=0 (disable) or non 0 (enable)

Exit

Returns nothing

Description

The LCD screen window normally tracks screen output to keep the cursor in view. Use this function to enable or disable this Screen synchronisation. (For example, you could disable synchronisation if using standard software that moves the cursor to unimportant areas of the screen.)

The default state is enabled. This default is restored by a cold start. Users can also control synchronisation by using the HOT key.

Get/set diskette port (+)

Entry

AH=84H
AL=0 (get) or 1 (set) BH-drive (0=A:, 1=B:)
BL=port (0=COM1, non 0=COM2), if set

Exit

AL=port, if get

Description

Use this function to define which port to use for which diskette drive, or to read the current port assignment for the specified drive.

The default state is for both drive A to be addressed as COM2 and drive B to be addressed as COM1.

Set/restore power menu (APM) (+)

Entry

AH=8SH
AL=0 (restore) or 1 (set)
BL=user software interrupt, if set

Exit

Returns nothing

Description

If APM is selected, when the user changes batteries, a menu will normally be displayed to allow the operator to change the battery type, capacity, etc.

Use this function to disable the usual menu and execute your own routine instead so you can handle battery changes from within an application. This routine must be vectored via the user interrupt specified by BL.

The default state (restored by a cold start) is for the normal prompt.

Get/set screen attribute table (+)

Entry

AH=86H
AL=0 (get) or non 0 (set)
DS:SI=start of attribute table buffer
BX=1234H

Exit

AH-0
buffer contains attribute table, if get

Description

If you set AL to 0, this function will read the attribute table into the 256 byte buffer pointed to by DS:SI. If you set AL to non-zero, the attribute table will be set to the contents of this buffer.

The first entry in the table is for text attribute 00H, the second for text attribute 01H and so on. Whenever a character is displayed, its character dot data is XORed with the appropriate entry in the table and the result is displayed.

For example, if a character has the attribute 01H and entry 01H in the table is 00H, the character will be displayed as normal; if the entry is FFH, the character will be displayed as inverse video.

The dot data is XORed a row at a time, with the least significant bits of the entry. For example, if the entry is 0FH (00001111), the righthand four pixels of the character will be reversed. In 8x8 mode, the righthand half will be reversed, while in 8x6 mode the righthand two thirds will be reversed.

Users can also select the particular attribute combination to display as reversed, by using the HOT key. This will override changes made by using this function, but you can pre vent access by using INT 15 function 7BH.

Note that this function has no effect in graphics modes, since there are text attributes in graphics modes.

Set INT 72H vector (+)

Entry

AH=87H
DS:DX=new service routine

Exit

Returns nothing

Description

Use this function to execute your own service routine when an INT 72H is issued. The routine starts at the location specified by DS:DX.

NOTE: You must also use DOS INT 21H function 25H to update the interrupt vector directly.

INT 72H is issued for three different reasons;

1: when the machine is about to turn off
2: when the machine is about to reboot
3: when the machine is powering up, in resume mode

In situations 1 and 2, AH-0 on entry. In situation 3, AH=1 on entry. You can use this information to take the appropriate action eg to turn off unsupported internal or external hardware options (AH-0) or turn on and initialise them (AH=1).

The default state (restored by a cold start) is to use the normal INT 72H service routine.

Get power up keys (+)

Entry

AH=88H

Exit

AH-00H
BX=column bit pattern
CL=row bit pattern

Description

Use this function to read which keys are defined as power-up keys. For more details, see the description of function 71H.

Get/set power mode (+)

Entry

AH=89H
AL=0 (get) or 1 (set)
BL=mode (0-SPM, 1-APM), if set

Exit

AH=00H
BL=current mode ie 0 (SPM) or 1 (APM), if get

Description

Use this function to change or check the current power control mode. Mode 0 is Standard Power Mode (SPM) while mode 1 is Advanced Power Mode (APM). Note that users can also change the power control mode by using the HOT key, unless access is disabled by using function 7BH or 7CH.

For details of how operation differs in the two power modes, see Part 2.

The power mode is not altered by a cold start or other events in normal operation.

Control power input (SPM) (+)

Entry

AH=8AH
AL=0 (disable) or non0 (enable)
SI=1234H (password)
DI=5678H (password)

Exit

AH=00H (success) or 0FFH (failure)

Description

Use this function to enable or disable power input from the AC adaptor, when using Standard Power Mode (use function 7FH in Advanced Power Mode). When power input is disabled, the AC adaptor will not charge batteries or supply power to **Hunter 16**. This should be used if Alkaline batteries are fitted.

Note that for this function to operate correctly you must also set SI and DI as indicated.

The default, restored on a cold start, is to enable power input.

Set charger temperature override (+)

Entry

AH=8BH
AL=0 (get) or non 0 (set) BH=maximum temp, if set
BL=minimum temp, if set

Exit

AH=00H (success) or 0FFH (failure)
BH=maximum temp, if get
BL=minimum temp, if get

Description

Use this function to set or check the temperature range within which charging is allowed. BH and BL specify the temperature range and must be in the range 0..7FH, with 0 being -20 degrees Centigrade and each increment being 0.63 degrees. For example, 28H is 5 degrees Centigrade.

The default, restored on a cold start, is to allow charging between 5 and 40 degrees Centigrade.

Get/set power save entry flag (+)

Entry

AH=8CH
AL=0 (get) or non 0 (set) BX=mask, if set

Exit

BX=current mask, if get

Description

Hunter 16 normally enters power save mode to conserve power whenever possible, for example, if an application is simply scanning the keyboard (waiting for input). However, it will not do so if an application is repeatedly accessing other BIOS services since this indicates that it is doing more than waiting for user input.

To allow entry to power save mode even while such accesses are being made, use this function. Set bits in BX to 0 to indicate which services should not prevent power save entry, as follows:

Bit	Service
0	INT 10H (screen services)
1	INT 13H (disk services)
2	INT 14H (comms services)
3	INT 15H (BIOS extension services)
4	INT 1AH (time/date services)
5	INT 21H (DOS services)
6	Direct write to video RAM
7	Access to 8250 UART
8	Access to 8253 (sound)

The default mask (FFFFH) is restored on a cold start, to prevent power save if any of these services is being used.

Get/set boost charge (SPM) (+)

Entry

AH=8DH
AL=0 (get) or non 0 (set)
BH=0 (prevent) or non 0 (allow)

Exit

AL=0 (get) or non 0 (set)
BH=0 (prevent) or non 0 (allow)

Description

In Standard Power Mode (SPM), charging is normally at 70mA, but you can use this function to allow a fast (200mA) charging rate for 1 hour, or check if fast charging is allowed. (In Advanced Power Mode this function is ignored; charging is fast until the battery level reaches 100%.)

NOTE: You cannot select fast charging within 6-7 hours of the previous fast charge.

The default, restored on a cold start, is to disable fast charging.